

Drone Arm Structural Analysis & Factor of Safety (FoS)

1 Geometric Parameters and Loading Conditions

The geometric parameters extracted from the CAD model and the operating conditions are as follows:

- **Length between centers (L):** 90 mm = 0.09 m
- **Cross-sectional width (b):** 30 mm = 0.03 m
- **Cross-sectional thickness (h):** 10 mm = 0.01 m
- **Motor thrust mass (m):** 1.0 kg
- **Gravitational acceleration (g):** 9.81 m/s²

2 Force and Bending Moment Calculations

2.1 Applied Thrust Force (F)

The upward force exerted by the motor at the tip of the arm is:

$$F = m \cdot g = 1.0 \text{ kg} \times 9.81 \text{ m/s}^2 = 9.81 \text{ N} \quad (1)$$

2.2 Maximum Bending Moment (M)

Treating the arm as a cantilever beam fixed at the frame attachment point, the maximum bending moment occurs at the root:

$$M = F \cdot L = 9.81 \text{ N} \times 0.09 \text{ m} = 0.8829 \text{ N} \cdot \text{m} \quad (2)$$

3 Cross-Sectional Properties

For a rectangular cross-section of width b and thickness h :

3.1 Distance to Outer Fiber (c)

$$c = \frac{h}{2} = \frac{0.01 \text{ m}}{2} = 0.005 \text{ m} \quad (3)$$

3.2 Area Moment of Inertia (I)

$$I = \frac{b \cdot h^3}{12} = \frac{0.03 \text{ m} \times (0.01 \text{ m})^3}{12} = 2.50 \times 10^{-9} \text{ m}^4 \quad (4)$$

4 Maximum Bending Stress Calculations

Using the elastic flexure formula:

$$\sigma_{\max} = \frac{M \cdot c}{I} \quad (5)$$

Substituting the calculated values into the equation:

$$\sigma_{\max} = \frac{0.8829 \text{ N} \cdot \text{m} \times 0.005 \text{ m}}{2.50 \times 10^{-9} \text{ m}^4} = 1,765,800 \text{ Pa} \approx 1.7658 \text{ MPa} \quad (6)$$

Alternatively, using the direct cantilever rectangular formula:

$$\sigma_{\max} = \frac{6 \cdot F \cdot L}{b \cdot h^2} = \frac{6 \times 9.81 \text{ N} \times 0.09 \text{ m}}{0.03 \text{ m} \times (0.01 \text{ m})^2} = 1,765,800 \text{ Pa} \approx 1.7658 \text{ MPa} \quad (7)$$

5 Factor of Safety (FoS) Analysis

The Factor of Safety is evaluated across various candidate materials using the material dataset properties:

$$\text{FoS} = \frac{\sigma_{\text{yield}}}{\sigma_{\max}} \quad (8)$$

Table 1: Material Properties and Factor of Safety Calculations ($h = 10 \text{ mm}$)

Material	Yield Strength (σ_{yield})	Bending Stress ($\sigma_{\max}$)	Factor of Safety (FoS)
Carbon Fiber Composite (CFRP)	600 MPa	1.7658 MPa	339.79
Fiberglass Composite (GFRP)	300 MPa	1.7658 MPa	169.89
Aluminum Alloy	275 MPa	1.7658 MPa	155.74
Wood (Birch)	80 MPa	1.7658 MPa	45.31
PLA Plastic	60 MPa	1.7658 MPa	33.98
ABS Plastic	40 MPa	1.7658 MPa	22.65